COMP3000: 6 parser → AssignmentOne

Specification

Your task is to create

- a Lox-style[1] parser for programs simulating water flows,
- three example programs,
- and a document explaining your language and parser.

The document is your chance to explain your decisions and implementation to your marker, writing a good one will help your marker give you grades for the other files as well. For example, you marker won't know how to compile your parser unless you tell them so you can't get parser marks unless you clearly describe that in your document. If your intention is not clear from the documents you submit, you won't be allocated grades, even for work that is ultimately correct. It is your responsibility to communicate clearly everything your marker needs in your submission. You marker will be one of the class teachers, so you can assume they are familiar with anything from class.

Note

There is ambiguity left in this description by design. You are aiming to make *the best possible domain specific language for waterflows* but we clarify a great deal to support grading your work. Imagination is needed to complete this task and experimentation is strongly encouraged.

Table A

To get you started, 20% of your marks are available just for completing the following table and including it in your document. It will help your markers follow along with your explanations.

[1] Please don't start a custom parser/interpreter from scratch or use any language other than Java as your implementation language. Your marker will need to be familiar with the basics of your approach - assume they are *only* familiar with the Lox parser.

What literal in your language represents a river that gets 10L/s of flow on the first day after 1mm of rainfall?	
What symbol in your language is used to show two rivers combine?	
Is the above symbol a "unary", "binary", or "literal"?	
What folder is the "working folder" to compile your parser?	
What command(s) will compile your parser?	
In your language, how long does it take all the water to work through a river system after 1 day of rain?	
Does your language include statements or is it an expression langauge?	
Which chapter of the book have you used as the starting point for your solution?	

A parser for water flows

We wish to develop a domain specific language for the *modelling* and *simulation* of water flows in river systems. A river system is a series of rivers. Each river is either

1. A root river or
2. The combination of the output of multiple other rivers or
3. The last river in the system is identified as the *output river*.

A *river system* is a series of connected rivers. Your domain specific language must be capable of representing a river system. Executing a program in this domain specific langauge (which we *don't* do yet) will compute the flows in every river at each day given a certain rainfall. For now you can ignore the complication of representing the rainfall and focus on a language which is capable of describing the river itself.

Example river system

You are free to build your examples from any real or imagined river system. Here I have described one example river system which makes a relatively easy example to show off all the features of your language which you can choose to use.

Figure 1 shows a river system situated near Canberra. The following statements are true of this river system:

- The root rivers in this system are googong, jerrabombarra, and uppper molongolo.
- Googong drains to dam 1
- Dam 1 drains to queanbeyan
- Central molongolo is the combination of queanbeyan, upper molongolo, and jerrabombarra
- Central molongolo drains to dam 2

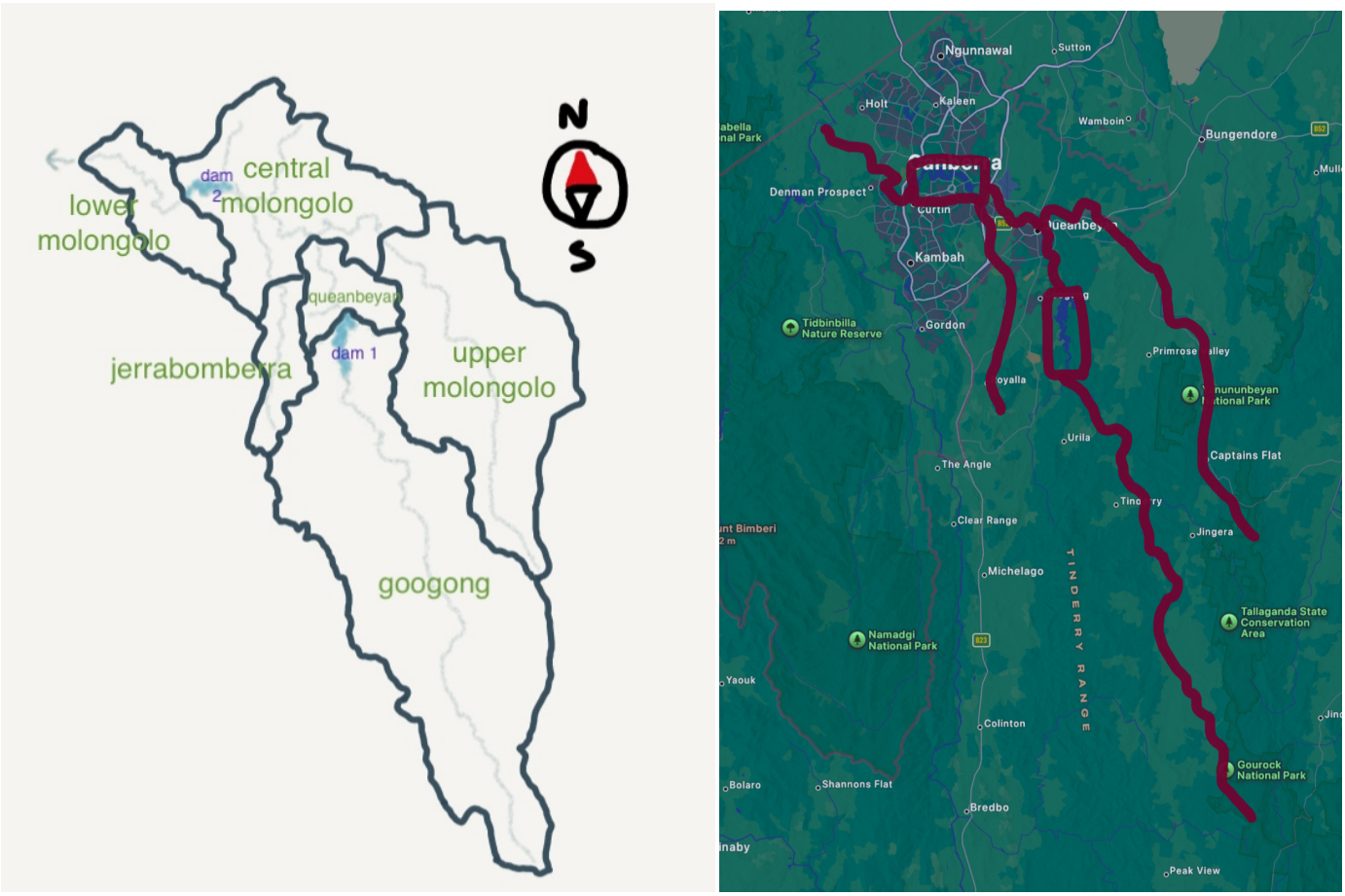

Figure 1: An example river system. On the left is a stylised map and on the right is all the draining rivers and dams on a map to help situate the river system.

- Dam 2 drains to lower molongolo[2]
- The output of the entire river system is the output of lower molongolo

We make the following simpliying assumptions:

1. The rainfall across a whole *river system* can be described with one number (in mm).
2. The rainfall across the whole river system is provided at run time by "magic". I.e. we can just assume for now that such a number exists and can be accessed.
3. The flow into one river is the sum of the flows out of its feeding rivers.
4. The flow out of a river is the flow in plus the water that has made it way into the river on that day.
5. All water that gets into a river will also flow out of it.
6. All water that falls from the sky will (within 10 days) make its way into the river.
7. The outflow of a dam is exactly the inflow to the dam[3].
8. Rivers never split into two or more rivers.

Use of Generative AI

An ability to use Generative AI is not a learning outcome of this course but GenAI is a tool you may choose to use along your journey. If your submission is primarily AI generated, you will likely find yourself in the bottom of each rubric item. Lox and Nystrom's notation are not well represented in AI training sets and Gen AIs are generally bad at language design. We value, and give grades for, your contribution to the solution since it is you (not the GenAI vendor) whos is earning the grade.

Note

You are free to use any and everything from your team's work on this task. You may not use other team's work directly or solicit for solutions. These remain contraventions of academic integrity. The work of the team belongs to all team members, so they can submit that without contravening academic integrity. However, this is your own submission. You can, and should, improve upon your team's work as much as possible. We have included a "similarity/creativity" score in the rubric so we can reward solutions that go beyond what was done in class. Note that *the document must be your own work*. It can heavily reference the team's work, but you should write it from scratch yourself.
You are also free to use any of the example solutions given by unit staff without referencing them.

[2] FYI: Dam 2 dams lake Burly Griffith in Canberra, you may have been there!
[3] We will relax this assumption later and have great fun controlling water flow with dams

Rubric

Rubric item	Fail (0)	Pass (50)	Credit (70)	Distinction (100)
Complete Table A (20%)	None answers accurately	2 or more answered accurately	6 or more answered accurately	All answered accurately
Grammar given in the document in Nystrom's notation (20%)	No grammar or substantially broken grammar in the document	A grammar with small errors	A grammar that can model rivers systems	A grammer that can model river systems and an explanation of how each example program parses according the grammar
Three example programs (20%)	No program or the programs don't model river system correctly	Three programs of some form	*pass* and the example programs all match the given grammar	*credit* and the Lox-like parser can actually parse the programs
Parser written in Java based on Lox codebase (20%)	None or not in Java	Code is based on the Lox from chapter 5 of text	Code is based on the Lox from chapter 5 and your additions fit with the style	If there is any improvement to the Chapter 5 Lox code with an explanation in the document, you will get this grade.
Uniquness and Creativity (20%)	A direct submission of in-class work	minor modifications to in-class work	substantial improvement on in-class work	a unique submission showing creativity

Examples for Similarity Rubric

Below I give total contribution tot he final assignment grade for certain scenarios:

0% You have been working with your team all semester and you take the solutions your team came up with, add *your own* document and submit it as-is.

14% You have been working with your team all semester and they also met outside class hour to try out other things. You take the solutions your team came up with, add *your own* document and submit it as-is.

20% You have been working with your team all semester. You take the solutions your team came up with and make some adjustments of your own to make it even better. You add *your own* document and submit.

20% You have been the main person doing all the work for your team and you submit that work, with a document of your own.

Answer

Still to do